

Zhecheng Yu

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Experiences

- **AI Researcher (Internship)** – KnowinAI, Shenzhen, China Apr 2026 - present
Embodied Perception Group
- **Visiting Scholar** – National University of Singapore, Singapore Aug 2025 – Apr 2026
NUS Ubicomp Lab PI: Prof. Brian Y. Lim
- **BEng** – Southeast University, Nanjing, China Aug 2022 – Jun 2026 (expected)
Training Base for Top Students in Computer Science (Honors Program) Avg. Score: 89.29

Research Interests

Computer Vision, Robotics, Explainable AI

Publications

- [1] **Learning from Human Gaze: Human-like Robot Social Navigation in Dense Crowds** AAAI 2026
Zhecheng Yu, Yan Lyu, Chen Yang, Tao Chen, Yishuang Zhang, Bo Ling, Peng Wang, Guanyu Gao, Weiwei Wu, Brian Y. Lim

Projects

- Generalizable Pick & Place Pose Prediction** Apr 2026 – present
Embodied Perception Group, KnowinAI

- Develop a VLA policy to predict 10-DoF gripper poses for picking and placing diverse objects across diverse scenarios.

- Explainable Visual Abduction** Aug 2025 – present
Collaborator: Prof. Brian Y. Lim, NUS

- Generate visually-grounded hypotheses instead of purely textual guesses to enable perceptual-level reasoning.
- Implement a selective abduction loop to eliminate inconsistent explanations and identify the most plausible cause.

- Gaze-Augmented AI** Jun 2024 – Jul 2025
Collaborator: Prof. Yan Lyu, SEU & Prof. Brian Y. Lim, NUS

- **From Gaze to Human-like Robot Crowd Navigation** Nov 2024 – Jul 2025
 - Collected an egocentric dataset featuring synchronized human gaze, video, and trajectory data in real-world crowds.
 - Developed a modular framework that predicts human-like gaze to identify socially saliency and guide navigation.
 - Demonstrated that incorporating human gaze improves both navigation performance and human alignment.
- **VR-Based Gaze Projection onto RGB-D Videos** Jun 2024 – Nov 2024
 - Developed eye tracking and gaze interaction functionalities in a virtual environment with a MR headset and Unity.
 - Integrate the MR headset with a depth camera into a multimodal data collection tool for real-world eye tracking.

Extracurricular Experiences

- Head of the Organizational Department** *The CSE Students' Union, Southeast University* Sep 2023 – Jun 2024
- Planned, organized and hosted large-scale campus events
 - Demonstrated strong leadership and teamwork skills

Awards

- **Outstanding Research-based Final Year Project (FYP)** (only 24 undergraduates university-wide) Dec 2025
- **China-Singapore SIP International Scholarship** 10,000 CNY (only 1 undergraduate department-wise) Oct 2025
- **National Grand Prize** 2024 National English Competition for College Students (top 1‰) May 2024